MultiMap: AI-Assisted Multi-View Streetscape Annotation

ANONYMOUS AUTHOR(S)

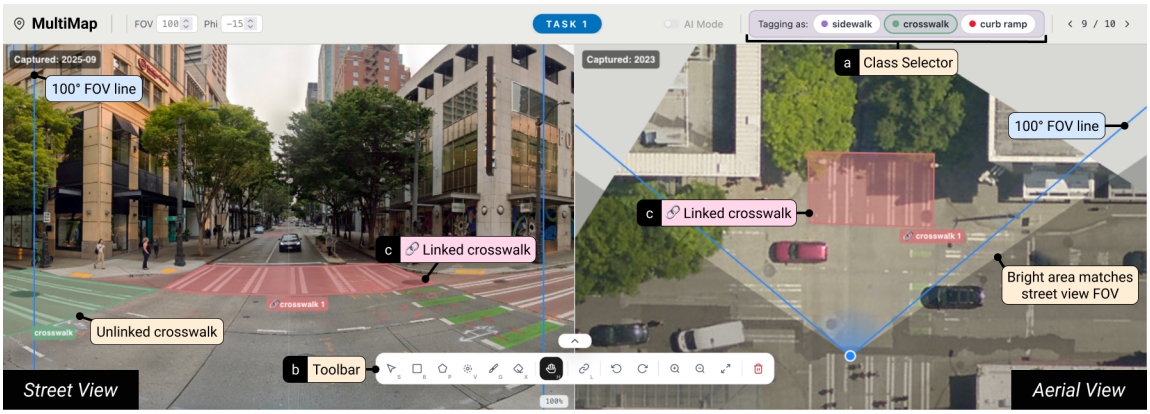


Fig. 1. We introduce MultiMap, a multi-view, AI-assisted streetscape annotation system that enables the labeling and linking of objects across multiple street perspectives. Users select the class they are tagging with (a) the class selector, draw and edit labels with tools from (b) the toolbar, and (c) link across the two views to create linked objects. Refer to video demo for further details.

Multi-view mapping tools combine the benefits of aerial imagery with streetscape photos (*e.g.*., from Mapillary, Google Street View) to help annotators accurately locate objects. The multiple viewpoints enable mappers to mitigate occlusion (*e.g.*, tree canopies) and leverage the affordances of each view (*e.g.*, roadways are easier to trace from top-down views but signage requires streetscape images). However, linking entities across views—the so-called *correspondence problem*—is a challenging, high-cognitive task. In this paper, we introduce and study *MultiMap*, an AI-assisted labeling tool aimed at helping mappers more quickly and accurately label street scenes and create correspondence between views. We evaluate MultiMap with 16 users, finding multiple views help users infer object positions; however, labeling with AI assistance was more accurate but, surprisingly, took significantly longer to create. We contribute findings on the challenges of streetscape annotation, a novel AI-powered detection pipeline, and design guidelines for future AI-assisted multi-view annotation systems.

CCS Concepts: • **Human-centered computing** → **Systems and tools for interaction design**; **Interactive systems and tools**.

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1 Introduction

Streetscape imagery has enabled fundamental advances in the scalable analysis and understanding of city streets: aerial views allow us to catalog sidewalks [24], green space [25], and parking [26], and street-level views enable us to detect signage [40], building facades [53], and curb ramps [44]. However, these works only focus on *single-modal* imagery, imagery taken from a single perspective (*e.g.*, street-view *or* aerial). Work that has used *multimodal* imagery has shown improved performance at characterizing buildings [39, 54], trees [55], and bike infrastructure [42], due to multiple image views naturally providing more information about a scene. This idea is powerful: combining multiple datasets and modalities (*e.g.*, through time, taken with handheld devices or through Google Street View, or through satellite or flyover imagery) has massive potential to expand our knowledge of the scene beyond what is available in just one perspective. However, despite this potential value, there is far less work that leverages multi-view imagery, in part due to a *multimodal correspondence problem*: we do not know which objects (or pixels) in one image map to which objects in the other [9, 39], and datasets that contain this information are few and limited to a single class or label type [39, 58]. As such, we lack the data to both train and evaluate models on their multimodal reasoning ability, and there is no standardized approach to creating these cross-view streetscape datasets.

To that end, we designed and developed MultiMap, a tool for labeling and linking streetscape infrastructure in side-by-side street level and aerial imagery. We use MultiMap as a platform to explore what challenges both experts and non-experts in mapping and Geographic Information Systems (GIS) face in multimodal streetscape labeling. Additionally, we implement an AI assistance mode (powered by a state-of-the-art foundation model) that proposes candidate labels and links, capitalizing on the increasing capability of zero-shot multimodal large language models (MLLMs), letting us explore the benefits and drawbacks of an AI-proposes, human-corrects labeling system. Our work is guided by the following research questions:

- **RQ1**: What challenges are inherent in street/bird's-eye view cross-modal labeling?
- **RQ2**: What design guidelines/principles aid cross-modal street-view labeling?
- **RQ3**: What automation strategies aid cross-modal labeling?

We investigated these research questions with a user study of 16 participants: eight with little to no mapping or image annotation experience and eight mapping and GIS experts, using MultiMap with and without AI Mode in a counterbalanced within-subjects study. We found that occlusion and temporal and structural differences in the images remain obstacles for streetscape labeling, but also that presenting two views of the same scene allowed users to anchor their inferences and labels based on what is visible in *both* images, rather than just one. Regarding AI assistance, users took *longer* to label with the AI, indicating that the time to review and edit a proposed label was longer than to make one from scratch; however, labels with AI assistance on were also more *accurate*, raising questions about the potential benefits and drawbacks of an AI-assisted labeling system.

In summary, we contribute **(1) formative findings** on the challenges inherent in multimodal streetscape image annotation, **(2) design guidelines** for future cross-modal annotation systems, **(3) a novel AI-powered system** for automatically generating cross-modal streetscape infrastructure label candidates, and **(4) formative findings** on the benefits and drawbacks of such an AI-assisted labeling system. We will release MultiMap's code for open science upon acceptance.

2 Related Work

We situate our work by synthesizing cross-disciplinary research on cross-modal streetscape models, datasets, cognitive science, and image annotation systems.

2.1 CV for Cross-modal Correspondence

Cross-modal correspondence refers to taking multiple images of the same scene and determining which pixels refer to the same object [9, 39]; this allows synthesis of information across multiple views into one shared understanding, increasing our ability to reason about a scene and reducing the drawbacks felt by one individual modality. For example, tagged datasets like UrBench [58] and OmniCity [39] match buildings across street-level and aerial views, allowing detection of footprint and land use from the aerial image while gathering data on building facades, height, and age from street-level [57]. Beyond buildings, Wegner *et al.* curate a cross-view tree dataset to accurately detect tree locations in Pasadena [55], and Lieu *et al.* use street-view and aerial images to detect bike lanes and types, and find performance improves when integrating aerial imagery [42]. Despite the demonstrated utility of multimodally aware CV, the above work only offers a single feature (*e.g.*, trees, buildings annotated using OpenStreetMap [39, 46, 54]), is limited to one type of label (*e.g.*, bounding boxes or classification), and there is little work exploring ways to create these multimodal datasets. These works demonstrate the need for (1) larger scale and more generalizable cross-modally annotated image datasets, and (2) guidelines and software to facilitate their creation, as our work provides.

2.2 CV on Streetscapes

Despite the improving ability of computer vision (CV) at analyzing and understanding urban infrastructure and streetscapes [24, 50, 53], computer generated labels have not reached parity with human ability when it comes to generating labels. Long-used architectures in deep learning that have been trained on task-specific data, such as convolutional neural networks (CNNs), have been used to detect sidewalks [24], land use [8], green space [25], parking [26], signage [16, 40], buildings [53], and curb ramps [44], but their utility comes in *scalable* streetscape understanding rather than perfect accuracy (for example, Hwang *et al.* found that when comparing two humans *vs.* one human and a model, the model had larger variance in estimating parking space width from aerial imagery [26]). This is all the more relevant as MLLMs like ChatGPT [48] and Gemini [5] are able to make zero-shot inferences across a wide range of objects and domains [11, 20, 29, 41, 50, 54, 56], in theory lending themselves well to the creation of new annotated datasets where previously trained models do not exist. However, they also have not yet shown ability to produce highly accurate labels; for example, Jiang and Hwang *et al.* showed an MLLM only achieved an F1-score of 0.83 [28] when detecting urban artwork from street-view imagery, and Han *et al.* found the intersection-over-union (IoU) of the predicted labels to the truth was only 0.42 in the best case [20]. Vision Banana, a state-of-the-art generative model, when tested on the Cityscapes dataset only achieved a .699 IoU [18]. As such, despite increasingly accurate and capable CV methods, human oversight is still required to produce the highest quality labels, and we explore how to facilitate that for multi-view streetscape data.

2.3 Human Cross-modal Perception

MultiMap inherently relies on a human's ability to synthesize multiple viewpoints for spatial understanding (*e.g.*, a top-down view with a first-person street level view). To understand how best to design visual cues to aid cognitive processing, we draw on cognitive science and mapping research [30, 33, 49]. Prior work suggests that cross-view correspondence

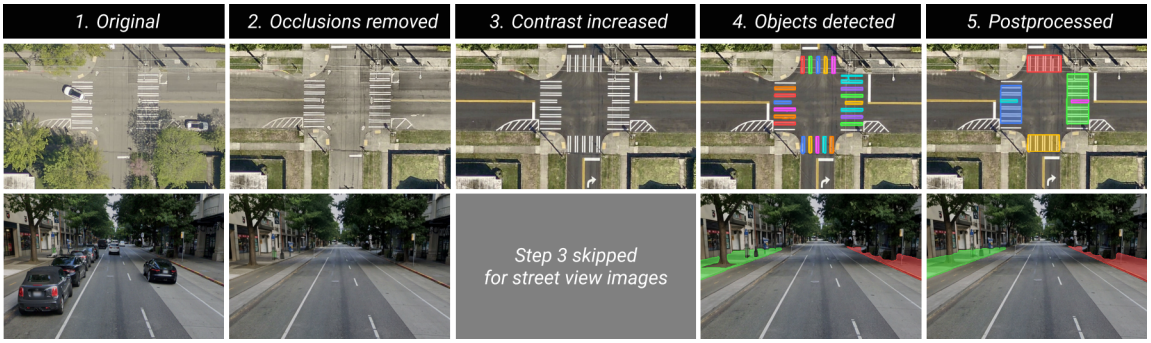


Fig. 2. Our AI candidate label pipeline has five stages. (1) Images are sampled from Tile2Net [24] and Google Street View [6], (2) Nano Banana 2 [7] is prompted to remove trees, shadows, and cars, and (3) sharpen paint and road markings. Then, (4) Nano Banana 2 is prompted to detect sidewalks, crosswalks, and curb ramps [18] (RampNet [44] for curb ramps in street-view), and (5) we postprocess the resultant labels to combine fragments and fill in alcoves. Labels are overlaid onto the original images in MultiMap.

relies on *spatial perspective taking*, the ability to imagine how a scene appears from another viewpoint [22, 33]. In mapping and wayfinding, Kiefer *et al.* found that humans who rely on landmark features and map symbols performed better than those who did not [30]; this is paralleled in a study by Skog *et al.*, who found that rotation did not influence ability to match houses in ground-level and aerial photos, suggesting that feature-level comparison is more influential [49].

Humans are also generally skilled at cross-view synthesis, as compared to CV models [19, 58]. CVBench investigated humans' and MLLMs' ability to answer questions that require synthesizing multiple camera views, finding that humans scored ~94% correct while the best MLLM scored under 40% [19]. Zhou *et al.* found that humans scored 17.4% higher than the MLLM on average on cross-view tasks [58]. Given this, through MultiMap, we explore how to provide AI assistance in a way that facilitates efficient and accurate label creation for cross-view image streetscape datasets.

2.4 Annotation Systems and AI Assisted Image Labeling

MultiMap builds on the extensive body of work in manual image annotation [2–4]. Commercial image labeling platforms such as Supervisely [2], Roboflow [3], CVAT [1], and LabelStudio [4] all feature point, polygon, and bounding box tools for annotating objects of multiple classes—features which MultiMap implements. Supervisely offers the ability to label and link the same object across multiple images [2], featuring synchronous image panning. Additionally, the OpenStreetMap editor [45] shows a street-level panorama while tagging objects in the aerial view, and synchronously shows the street-view heading on the aerial map. MultiMap builds on top of these features to tailor to labeling multiple views in the streetscape domain, further exploring the interaction design of multi-view annotation.

Due to the tedious and context-specific nature of manual image annotation [15, 35], researchers and systems have explored AI assistance to improve speed and accuracy [2, 3, 13, 36, 38]. Many commercial tools generate polygon labels from segmentation model (*e.g.*, Segment Anything [31]) output on-click. Research has explored different ways to provide AI input: C3Det learns from human labels and automatically detects (with bounding boxes) similar instances [36]; MONAI Label presents segmentation labels of radiology scans that users can then edit [13]; LabelAid surfaces a popup when it detects the user has potentially made a mistake [38]. C3Det, MONAI, and LabelAid all found that with AI assistance, labeling is faster or requires less effort [13, 36, 38]. One system (HEPHA [59]), in which users edit AI

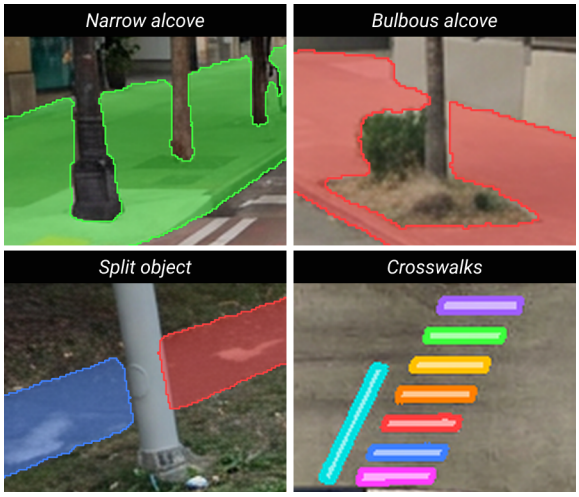


Fig. 3. We observe four types of common errors in AI-generated labels: narrow and bulbous alcoves, split objects, and fragmented crosswalks. We implement postprocessing steps to fill in alcoves, bridge split gaps, and combine fragmented crosswalks (Section 3.2.3).

proposed labeling *rules* rather than labels directly, found that users took longer but the resulting classification labels were more accurate [59]. However, despite existing work on AI assistance for single-view images, few have explored *multi-view* image labeling, as MultiMap does. ARGUSTRACK combines multiple views to a single view for labeling, then reprojects to the original camera viewpoints. However, the camera positions are relatively similar and not as disjointed as street-level *vs.* bird's-eye-view (BEV) [52] as in MultiMap. As such, in MultiMap, we explore extending these single-modal AI techniques to a multi-view labeling system, and further implement an algorithm for identifying and suggesting when two objects across different views are the same.

3 The MultiMap System

Grounded in prior literature, we built MultiMap: a multi-view streetscape annotation tool to tag and link objects across street and aerial views, with AI generated label suggestions powered by a state-of-the-art foundation model.

3.1 Data Sources

We present images in MultiMap as pairs: one street-level image to one aerial image, matched by latitude and longitude. For street-level imagery, we use Google Street View (GSV) [6] for its high-quality and comprehensive street-level coverage [27, 53]. For aerial BEV imagery, we use a high-resolution open-source orthographic image dataset provided by King County Seattle [51] captured in 2021, downloaded using Tile2Net [24]. We sample 256x256 tiles at zoom-level 20 (~15 cm/pixel), and center each tile on the reported latitude and longitude of each GSV panorama.

3.2 AI Candidate Label Generation

To provide AI label suggestions, we implement a novel pipeline that produces segmentation labels of sidewalks and crosswalks, and point labels for curb ramps. Where possible, we use a state-of-the-art MLLM (Nano Banana 2 [7]) for labels, to reflect a context where an MLLM is used to generate imperfect labels, zero-shot, in a new context (as

in [20, 28]). All image processing and detection is preprocessed before uploading to MultiMap; there is no real-time inference. See Figure 2 for the pipeline, and the Appendix (Section A.1) for full prompts.

3.2.1 Preprocessing. Occlusion, shadows, and lack of contrast are major challenges for streetscape image analysis [26]. As such, before object detection, we run a "cleanup" step: for aerial images, we prompt AI to *"remove trees, shadows, and cars occluding the road and sidewalks,"* and for street-view images we prompt Nano Banana 2 to *"remove cars and pedestrians."* To boost paint visibility (which can be worse in aerial imagery due to lower resolution) we then run a second pass on the aerial images to *"make road markings more clear."* In all prompts we specify not to add or change existing geometry, only to "fill in" objects that are already there. We then pass the "cleaned" images on for object detection. See Figure 2 steps 2 & 3.

3.2.2 Detecting Objects. Inspired by [20], we use Nano Banana 2 [7] to detect sidewalks, crosswalks, and curb ramps. We borrow heavily from the Vision Banana approach [18] and prompt the model to *"generate a semantic segmentation visualization of this image, using this color mapping: {"sidewalk": "green", "road": "red", "crosswalks": "blue", "buildings": purple}."* For curb ramp point labels, we prompt it to *"draw a circular marker on top of each curb ramp in this image."* Crucially, these labels are ultimately laid on top of the *original*, non-preprocessed images, so generative artifacts or hallucinations are not seen by the end user. Detecting curb ramps from street-view images is the only mode where we do not use Nano Banana 2, as we found the model produced excessive false positives that were more distracting than helpful. As such, we use RampNet [44], a state-of-the-art model specifically fine-tuned for curb ramp detection in street-view images. See Figure 2 step 4. To evaluate detections, we use the same approach as when evaluating the participants' labels; see Section 4 for method and results. For full prompts see Section A.1.

3.2.3 Postprocessing. We run postprocessing steps (Figure 2 step 5) on the crosswalk and sidewalk labels, as we noticed four common error types in the generated labels: narrow and bulbous alcoves, split objects, and crosswalk stripes being detected as individual objects (see Figure 3). For street-level imagery, all sidewalk steps run in a row-rescaled image space that approximately undoes perspective foreshortening, and all distance thresholds are expressed as fractions of the typical sidewalk width per-image. All thresholds below were learned through experimental iteration.

Alcoves Alcoves occur when an object (*e.g.*, tree trunk, fire hydrant, signpost) occludes the sidewalk. These come in two types: narrow alcoves, where the alcove is roughly the same width throughout, and bulbous alcoves, where the alcove is cave-like with a narrow opening. To fill in narrow alcoves, we apply a morphological closing [47] to each label with an elliptical structuring element 0.6 sidewalk-widths in diameter. A candidate fill is only accepted if its area is at most 5% of the label and more than half of it borders the existing label. This does not work on bulbous alcoves, as the fill only fills concavities smaller than its structuring element (whereas the bulbous alcove has a much larger interior). As such, we take a reachability approach: first, as tree trunks are often the cause of such alcoves, we detect tree trunks using SAM3 [31]. Then, considering sidewalk labels and tree trunks as walls, any area that is not reachable by an object 0.8 sidewalk-widths wide is considered an alcove, and filled in from the center.

Split Objects Split objects occur when an object, like a tree trunk, cuts the label into multiple pieces that, on inspection, should clearly be the same object. To merge split detections, we first collect all the pixels within 1.4× the smallest distance between the two labels—for a split object, this will take the shape of a rectangular strip. We then merge the two labels (and fill in with the pixels) if (1) the two labels are closer than 0.6 sidewalk-widths, (2) the strip is at least three times longer than its own width, (3) the strip is less than half the area of the smaller of the two labels, and (4) if the strip and the two labels form a contiguous region. As many of the splitting objects are tree trunks that result

Table 1. We score precision, recall, F1, and IoU of labels between ground truth created by one human researcher (R1), against a second researcher (R2)'s labels, the AI pipeline without postprocessing, and the AI pipeline with postprocessing. Curb ramp labels are points and therefore do not have an IoU score. Postprocessing measurably improves the AI labels. Neither AI condition nears R2's agreement with R1.

	R2				AI w/o postprocessing				AI w/ postprocessing			
Class	P	R	F1	IoU	P	R	F1	IoU	P	R	F1	IoU
Sidewalk	1.00	1.00	1.00	0.76	0.73	0.96	0.83	0.61	0.78	0.96	0.86	0.62
Crosswalk	1.00	0.99	0.99	0.80	0.40	0.98	0.57	0.52	0.57	0.98	0.72	0.66
Curb Ramps	0.99	0.91	0.95	N/A	0.57	0.77	0.66	N/A	0.57	0.77	0.66	N/A
Total	1.00	0.97	0.98	0.77	0.59	0.91	0.72	0.59	0.67	0.91	0.77	0.63

in wider, more square-shaped strips, we apply threshold loosening across tree trunks (the same detected using SAM3). If the strip is composed entirely of tree trunk pixels, then the rest of the requirements are waived and it is filled in.

Disconnected Crosswalks AI generated labels often detect each individual crosswalk stripe instead of the entire crosswalk surface. To combine stripes, we first fit a minimum-area rectangle to each detected stripe, giving its axis, length, width, and two end points. When three stripes or more (1) have axes that agree within 13°, (2) have widths that are within 0.4x to 2.5x of each other, and (3) overlap by at least 50% along each other's axes, they are considered grouped. Then, we fit two RANSAC [17] lines along the short-edge axes of the group; any stripe that lies farther than 42px in street-view and 9px in aerial from either line is removed from the group. In street-view, as the ground-level perspective warping can influence the prior tests, if any fail we run a fallback test. For every pair of detected stripes, a line is drawn between their centers. Every other stripe's center's distance to that line is measured. If it is sufficiently close (within 0.6x of the stripe's diagonal length), it is added to the line. The line with the largest number of stripes is considered grouped. Finally, the group of crosswalk stripes is merged by taking the convex hull of all contained stripes. While this postprocessing step combines many disjointed crosswalk stripes, occlusion and perspective warping can still weaken its power—as such, human edits are still required.

We evaluate the performance with and without postprocessing, and find that postprocessing measurably improves labels but not to the level of a human (see Table 1). As such, a human annotator is still required to bring the AI generated labels to human quality.

3.2.4 Inferring Links. We implement a coarse, position-based matching system to produce candidate link pairs between the two views. We measure each annotation's centroid's angular position in space relative to forward facing in the street-view panorama. We then propose a candidate link when the two objects of the same class are within 25° of each other. As an annotation in one image may have many candidates in the other, MultiMap surfaces all for the user to choose between. This algorithm is limited to position; it does not consider appearance, size, or shape.

3.3 User Interface Design and Implementation

Informed by prior literature and existing tools, we iteratively created *MultiMap*, a novel multi-view streetscape annotation system that displays street-view and aerial imagery side by side and offers editable AI generated candidate labels. Below, we describe MultiMap's UI, interaction modes, and implementation.

3.3.1 Image Views and Presentation. MultiMap presents two images side-by-side for the user to label: a street-view panorama on the left, and an aerial view on the right (Figure 1). These images are taken at the same location, and we rotate the aerial image such that "up" points the same direction as "forward" in the street-view image (SVI). There is a

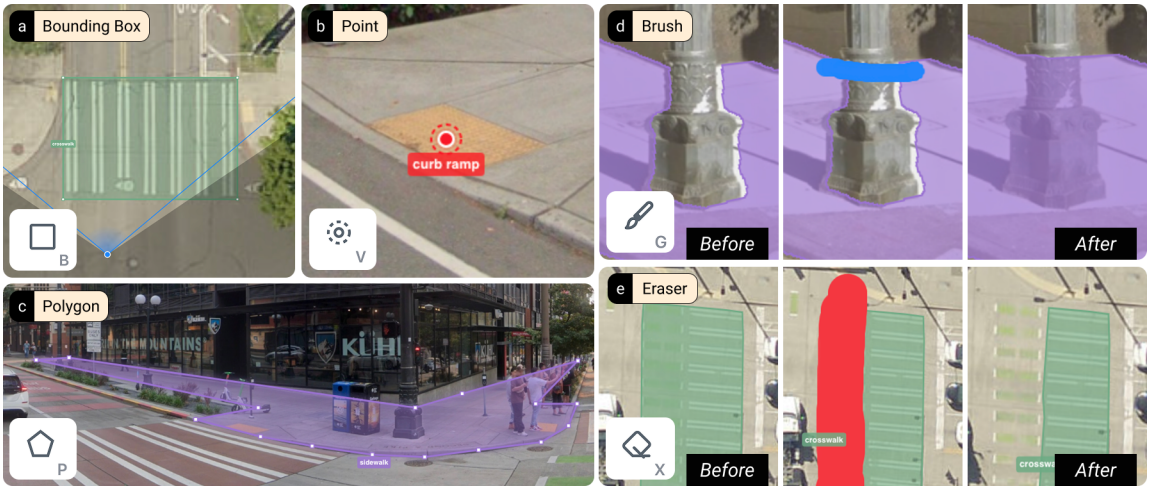


Fig. 4. MultiMap implements the following annotation tools: (a) bounding boxes, (b) points, and (c) arbitrary polygons, that mirror existing image annotation tools [2–4]. We additionally implement (d) a brush tool that adds area where drawn and merges annotations, and (e) an eraser that removes area and can split annotations.

toolbar along the bottom of the screen where the user can select tools, with a button to move it to the top of the screen (and *vice versa*) in case it is obstructing the image views.

By default, the street-view panorama is framed to show 112° of the panorama, centered on the crop's heading. Blue vertical bars mark 100° field-of-view (FOV) of the perspective crop, which is the region used for the study and analysis. The aerial view mirrors this, by showing blue lines in a 100° cone. Additionally, we dim the region of the aerial image outside of the visible FOV range of the street-view image. The blue cone and bars are fixed markers, whereas the dimmed region updates and rotates synchronously to match the pan and zoom on the street-view side. Users can pan around the image by selecting the "hand" tool (H), then clicking and dragging. Horizontal panning of the panorama is limited to 130°. Zooming is linked between the two views; scrolling with the mouse, or with the zoom buttons in the toolbar (+ and −) zooms both images by the same factor (zooming into cursor position when scrolling). Pressing "zoom-to-fit" (or the 0 key) resets both views to the 112° default.

3.3.2 Creating and Linking Labels. **Creating Labels** The user can create a label for a given class, by first selecting the class they are annotating in the top right corner (*"sidewalk," "crosswalk,"* or *"curb ramp"*). There are three different types of labels, each with their own tool and hotkey: bounding boxes (B), polygons (P), and points (V). These mirror the options in widely used labeling software like Roboflow [3], Supervisely [2], and LabelStudio [4]. Bounding boxes allow the user to draw a box around objects, the polygon tool allows users to create arbitrarily sized and shaped objects by placing points, and the point tool allows the user to place points by clicking (see Figure 4). When selecting the *"curb ramp"* class, the bounding box and polygon tools "gray out" and become unselectable, as curb ramps must be points. When changing to a different class, only the annotations for that class are visible.

Editing Labels Using the mouse tool (S), the user can move an annotation, or vertices, effectively editing the annotation. When selected, the user can also delete the annotation (by pressing the trash can button on the toolbar, or delete), or change the annotation's class.

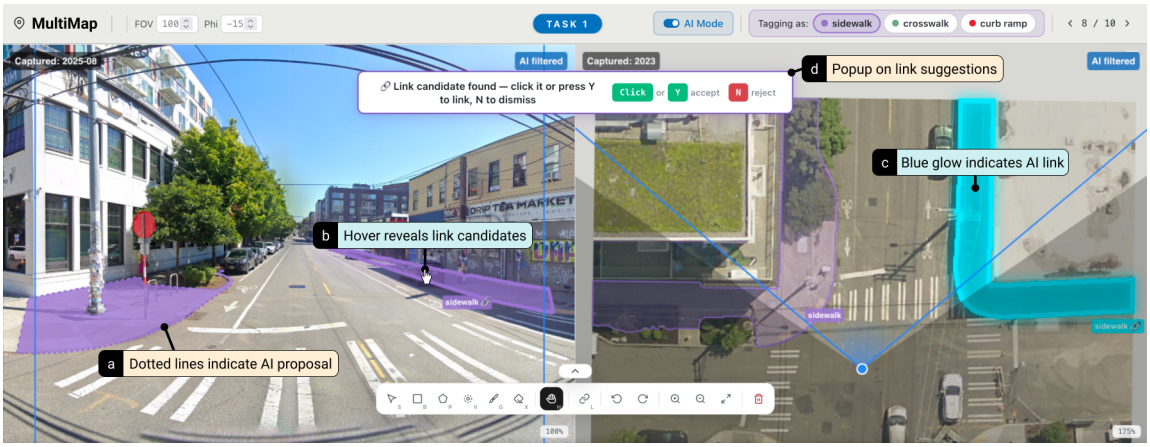


Fig. 5. In AI Mode, MultiMap proposes editable candidate labels and candidate links. (a) Proposed labels are shown with dotted borders, and users can accept or reject the label. (b) Hovering over an annotation reveals its link candidates in the other view, which (c) glow light cyan and pulse, and (d) a popup displays instructing the user to accept with Y or a click and reject with N. Using the brush or eraser on an unaccepted label accepts it automatically.

We also implement a *"brush"* and *"erase"* tool (Figure 4). Both are circular drawing tools, with modifiable diameter. The brush tool adds area to an existing polygon shape and merges multiple annotations that it crosses. When merging multiple shapes, we also include the space between them using either the (1) convex hull, or (2) the α-shape with α set to the larger of either the 2×brush radius or 1.15×the largest gap between merged shapes. We choose the convex hull if its area is less than 15% larger than the α-shape area. This allows crosswalk stripes (convex hull) to merge into a rectangle, and prevents L-shaped sidewalks (α-shape) from becoming a triangle. The erase tool erases where drawn, and can split one annotation into multiple.

Linking Labels Linking labels across image modalities is the core distinguishing feature of MultiMap. When selecting the *"link"* tool (L), if the user clicks an annotation on one image and then an annotation on the other, those two annotations will be linked and a chain icon will appear next to their names. Linked annotations are additionally denoted by sharing the same highlighted color, and a unique identifying number appears next to both names (*e.g.*, *"sidewalk 2"*). The linking tool remains active, so the user can link multiple pairs in succession. See Figure 6.

3.3.3 AI Assistance. MultiMap also includes *"AI Mode,"* which displays AI generated label candidates that the user can accept, edit, and reject. They appear as polygons with dotted borders, and when hovering over one a pop-up appears prompting the user to accept or reject the AI suggestion. The user can accept by clicking it or pressing Y, and they can reject it by pressing N. Using the brush or erase tool on any unaccepted label will automatically accept it.

AI Mode also suggests object links. When hovering over an annotation in one view, if the system detects a potential link in the other, that annotation will change color to a light cyan and "pulse"—additionally, a popup will appear stating *"link candidate found,"* giving the option to press Y or click to accept, and N to reject. If multiple links are detected, all will glow, and the user is prompted to click one to select.

3.3.4 Implementation. MultiMap is a static web app implemented with HTML, CSS, and JavaScript and uses JSZip [32] to create the participant's export directly in the browser. Image views are drawn with the HTML5 Canvas 2D API, which lets us control the coordinate transformations needed to keep the two views aligned while the user zooms and

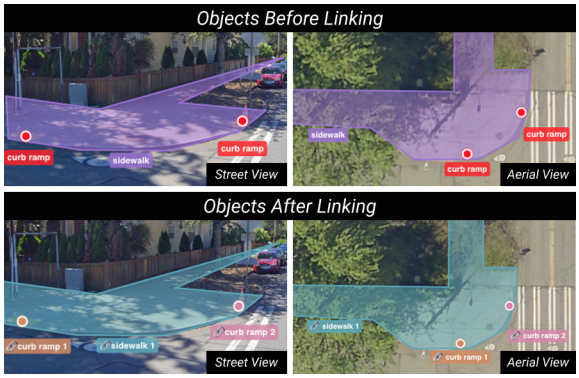


Fig. 6. Linking is a core, defining feature of MultiMap. With the link tool active, clicking an annotation in one view and then an annotation in the other links the pair. They then share a highlight color and the same identifying number. A chain icon also appears beside each name of the linked labels, indicating that they are linked.

pans. We preload all images and candidate labels to our web server for speed and consistency. Session state, annotations, and interaction logs are stored in the participant's browser (`localStorage`, with imagery cached in IndexedDB), and are written out only when the participant exports their annotations at the end of the session. Artificial intelligence was used to assist with implementation.

4 User Study

To explore the challenges in multimodal streetscape image annotation and the potential benefits of an AI-propose, human-correct image annotation system, we evaluate MultiMap with 16 participants: eight mapping and GIS experts and eight non-experts. Our study was deemed exempt by the Institutional Review Board (IRB) at our university.

4.1 Participants

We recruited 18 total participants (plus three pilot participants) through a combination of word-of-mouth, snowball sampling, and posting to a local mapping and GIS focused Discord channel. All participants filled out a pre-screener before participation that asked for basic demographic information, as well as their self-assessed expertise at *"GIS, mapping, or geospatial tools,"* using the five-point Dreyfus model of skill acquisition [14] ("*Novice*," "*Advanced Beginner*," "*Competent*," "*Proficient*," "*Expert*"). Participants who rated their expertise at "*Proficient*" or higher were marked as experts. Two participants declined to continue the study after completing the first task, and we do not include them in analysis. In total, this resulted in eight experts and eight non-experts. Participants were compensated US$25 per hour. See Table 2 for participant table.

4.2 Study Design

We conducted a 1×2 within-subjects study comparing annotation performance with and without AI assistance.

Image Selection and Sampling Measuring AI Mode's effect on labeling performance requires two distinct image sets (one labeled with AI Mode, one without), so that participants never label the same images twice. To prevent set differences from confounding the results, we design the sets to be roughly equivalent. We collected two sets of 10 images: *"Set A"* and *"Set B."* We first downloaded and sampled SVIs from GSV at 30 meter intervals across Seattle. From that pool, we randomly selected five images around intersections (containing crosswalks and curb ramps), and five additional

Table 2. We recruited eight experts and eight non-experts in GIS and mapping. Participants were assigned to one of four counterbalancing tracks (T1-4), with experts and non-experts sequenced independently. For self-rated expertise, we use the five-point Dreyfus model of skill acquisition [14] ("*Novice*," "*Advanced Beginner*," "*Competent*," "*Proficient*," "*Expert*"). Those rating themselves as *Proficient* or higher were treated as experts. Preferred mode is participants' response to what labeling mode (AI or Manual) they would choose to label 100 more image pairs. Three pilot participants are excluded from analysis.

P	Age	Self-rated GIS Expertise	Mouse or Trackpad	Track	Preferred Mode
Pilots					
Pilot1	18 to 24	Novice	Trackpad	T1	N/A
Pilot2	18 to 24	Novice	Trackpad	T2	N/A
Pilot3	18 to 24	Advanced Beginner	Mouse	T1	N/A
Non-Experts					
N1	18 to 24	Competent	Mouse	T1	Manual
N2	18 to 24	Novice	Mouse	T2	AI
N3	18 to 24	Novice	Mouse	T3	AI
N4	18 to 24	Novice	Trackpad	T4	Manual
N5	18 to 24	Novice	Trackpad	T1	Manual
N6	25 to 34	Advanced Beginner	Trackpad	T2	Manual
N7	25 to 34	Competent	Mouse	T3	Manual
N8	18 to 24	Novice	Trackpad	T4	AI
Experts					
E1	65 to 84	Proficient	Mouse	T1	Manual
E2	45 to 64	Expert	Mouse	T2	Manual
E3	35 to 44	Expert	Mouse	T3	AI
E4	25 to 34	Proficient	Trackpad	T4	AI (toggleable)
E5	25 to 34	Expert	Mouse	T1	AI (w/ better labels)
E6	18 to 24	Proficient	Mouse	T2	Manual
E7	35 to 44	Expert	Mouse	T3	AI
E8	25 to 34	Proficient	Mouse	T4	AI (depending on label quality)

images (containing sidewalks), constructing Set A. For each Set A image, we randomly sampled a Set B counterpart with the same *"road type"* (*e.g.*, "primary", "secondary") and sidewalk configuration (*e.g.*, *"sidewalk:both"*, *"sidewalk:left"*) according to the OpenStreetMap API [46], yielding 10 pairs. We manually inspected and visually confirmed that each pair had similar road structure (*e.g.*, lanes, sidewalks, crosswalks), environment (*e.g.*, vegetation, road quality, aesthetics), and infrastructure arrangement. We additionally counted crosswalks, curb ramps, and sidewalks per image so that totals of each feature were equal across sets. Pairs failing either check were resampled until they matched. Finally, we are left with Set A and Set B: two sets of 10 images each, that are equivalent in structure, aesthetics, and infrastructure counts (*crosswalks*=11, *curb ramps*=16, *sidewalks*=22).

Tasks and Counterbalancing Our study is a 1×2 within-subjects study; however, given two different image sets, to adjust for confounds we structure it as a two-condition within-subjects design, counterbalancing condition order and image-set assignment across four sequences: **(T1)** Manual Mode on Set A followed by AI Mode on Set B, **(T2)** AI Mode on Set A followed by Manual Mode on Set B, **(T3)** Manual Mode on Set B followed by AI Mode on Set A, **(T4)** AI Mode on Set B followed by Manual Mode on Set A. Participants were assigned to each track in sequential order of participation, with experts and non-experts in separate sequences (such that each track had two expert participants and two non-expert participants). We asked participants to label sidewalks, crosswalks, and curb ramps: three common streetscape infrastructure features that all target a different dimension of streetscape annotation. Sidewalks come in a

Table 3. Precision, recall, F1, and IoU for labels created in Manual Mode *vs.* labels created in AI Mode, compared to ground truth. Precision, recall, and F1 scores are largely uniform between the two conditions; IoU is slightly higher in AI Mode, suggesting labels created in AI Mode were more accurate.

	Manual Mode				AI Mode			
Class	P	R	F1	IoU	P	R	F1	IoU
Sidewalk	0.98	0.95	0.96	0.63	0.97	0.98	0.98	0.68
Crosswalk	0.95	0.99	0.97	0.77	0.94	0.99	0.97	0.78
Curb Ramps	0.88	0.89	0.88	N/A	0.88	0.89	0.89	N/A
Total	0.94	0.94	0.94	0.68	0.93	0.96	0.94	0.71

variety of shapes and often extend through the length of the image, requiring the polygon tool; crosswalks are blocky and rectangular, suitable for the bounding box tool; curb ramps are point objects with no definitive boundaries.

Questionnaires and Post-Task Questions After each task, users completed the NASA Raw-TLX [21], followed by open-ended questions targeting the user's thoughts and experiences (*e.g.*, *"which objects or features did you feel were easier to label?"*). Following AI Mode, users completed additional Likert-type questions, including if they found AI Mode useful (using the 7-point Perceived Usefulness scale from Davis' Technology Acceptance Model [12]), if they felt the AI's labels were accurate, and if they found the AI's link suggestions accurate. After completing both tasks, users completed the System Usability Scale (SUS) questionnaire [37] (sans the question "I think I would use this product frequently," as it confused our non-expert participants who do not regularly do GIS work), and we asked additional, open-ended questions to understand the user's holistic experience and comparisons between Manual *vs.* AI Mode. All Likert-type questions other than the NASA Raw-TLX and the SUS were 7-point. See the Supplementary Materials for the full protocol.

4.3 Apparatus

All studies were run virtually over Zoom, using a custom-built version of MultiMap with the images and AI labels pre-loaded, and the tracks (*e.g.*, T1, T2) coded in and selectable at the intro screen. Prior to each task (Manual and AI Mode) there was a tutorial, where the user was given five image pairs to practice on. Both tutorials (Manual and AI Mode) used the same five image pairs, the only difference being whether AI Mode was active. In all labeling tasks, participants were asked to label objects within a 100° cone facing forward (denoted by blue bars), and that while they could label outside of the bars, we would only analyze what is inside the bars. For each task, the user was responsible for clicking into the next image pair, then pressing the "Finish this set" button. Once a set is completed, the user is unable to return to the images. All Likert-type questionnaires were built into the system itself, such that after completing a task, the user completes them within the system rather than leaving the platform. Before continuing to the next step, the system reminds the user to pause and wait for the researcher to tell them to continue. All actions while interacting with the system are logged and saved to local storage, including number of annotations created, time spent per annotation, and tool usage. Once completing all tasks, the user is prompted to "Download my annotations", which exports all log and questionnaire data as a zip file.

4.4 Procedure

4.4.1 Introduction. Each session began with an introduction to the study, a high-level explanation and overview of the tasks and flow, and then informed consent. The participant was then asked two open-ended background-related questions to understand their experience with image annotation and map navigation.

4.4.2 Practice. To start with MultiMap, the user was first shown a pre-recorded video tutorial of MultiMap's manual mode introducing them to the system, layout, tools, and usage. Following that, the user was asked to share their screen and was sent a link to open MultiMap. After inputting a researcher-provided ID and track number, the user completed a tutorial where they were given five practice image pairs and told to *"get comfortable with drawing, editing and linking."* Upon completion of the Manual Mode tutorial, the user is shown a leaderboard displaying their time (total, and per class) and their position among previous participants. The user was then told that this leaderboard would similarly appear after completing each task. At this stage, if the user was to perform Manual Mode first, they moved onto the task itself. If the user was to perform AI Mode first, they were then shown an additional tutorial video explaining AI Mode, including accepting and rejecting labels and automated link suggestions, then practice on five image pairs, before moving onto the first task. Manual Mode-first participants were shown the AI Mode tutorial directly prior to starting their AI Mode task.

4.4.3 Task Flow. In each task, the user was asked to fully label the 10 image pairs, thinking aloud. After completing the set, they then completed the post-task questionnaire and the researcher asked them the open-ended post-task questions. They were then shown the task leaderboard before moving on to the next task. After completing both tasks, they then completed the post-study questionnaire on the system, then were asked to export their labels and send them to the study team via email. Finally, they stopped sharing their screen and were asked the post-study open-ended questions, followed by a final debrief, including providing compensation details.

4.5 Pilot Studies

Prior to the main study, we completed three pilot studies to inform system and study design. Following each, minor changes were made to the system, including only showing class annotations when that class is selected, adding the ability to move the toolbar, adding the ability to accept AI labels with the brush tool, and implementing the leaderboard. The largest change was halving the image pairs per task from 20 to 10: at 20, the full study took 3+ hours, causing fatigue and leading our second pilot participant to *"rush through the second task just to get it over with,"* invalidating any potential results. We do not include the pilot participants in the quantitative or qualitative analysis.

4.6 Analysis

We analyzed participants' quantitative performance across speed and accuracy, and their subjective ratings by comparing their NASA Raw-TLX responses between conditions.

4.6.1 Speed. For speed, we timed how long each user took to label, aggregated over class and task condition. Before analysis, we manually corrected the times of two participants who took a mid-session break, using the session recordings. The outcome measure for speed was time per task and per class, log transformed before analysis. Where residuals were normal (Shapiro–Wilk and Q–Q inspection), we fitted linear mixed models (LMMs) with mode (within: manual vs. AI), expertise (between), and their interaction as fixed effects, task order (first vs. second) and image set (A *vs.* B) as covariates, and participant as a random intercept: time ~ mode × expertise + order + set + (1 | participant). Where non-normal, we fitted a generalized linear mixed model (GLMM) with the same structure, modeling raw durations with a Gamma distribution and log link. For per-class statistical tests, we correct p-values with Holm's sequential Bonferroni procedure [23].

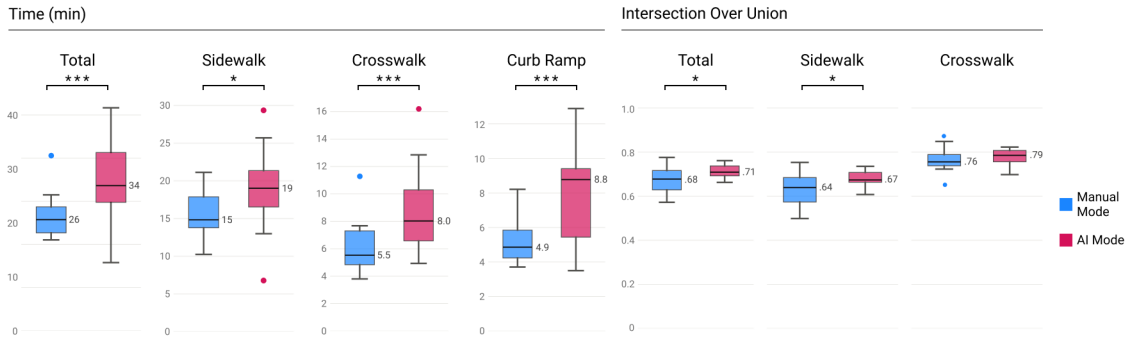


Fig. 7. Tukey box-and-whisker-plots showing differences in time and IoU between Manual Mode (in blue) and AI Mode (in pink). AI Mode was significantly slower across all classes. However, IoU was significantly more accurate for sidewalk annotations in AI Mode.

4.6.2 Accuracy. For accuracy, we compared user labels to a researcher-created ground truth (GT) set, computing precision, recall, and F1 for object identification and intersection-over-union (IoU) for label quality. To contextualize user performance, we also baselined our GT against a second researcher-created set and against the raw and postprocessed AI-proposed labels. Before analysis, we clipped all labels to the 100° forward-facing cone (blue bars).

To match labels between GT and the predicted set, we use a "fuzzy" matching algorithm. First, if any GT label envelops more than 70% of any user labels (or vice versa), they are considered matched and removed from both GT and user label pools. Second, we use the *Hungarian* method for bipartite matching [34] with an IoU exceeding 0.5, and again remove matched objects from the pools. Finally, we match based on distance: if any curb ramp label is within 185px in street-view or 45px in aerial of a GT label, they are considered matched, and if any crosswalk or sidewalk are touching, they are considered matched. We only match *within classes*, so if a user falsely labeled one object as the wrong class, that would not match to GT. This fuzzy approach allows for flexibility in matching labels that are fragmented, or barely overlapping but clearly denoting the same object (as can happen with thin objects like sidewalks). Any lenience in the matching stage would then reflect as a lower score in the IoU phase, as a poorly aligned object would have a lower IoU. Thresholds were learned through experimental iteration.

Our outcome measures for accuracy were the micro-F1 and per-class F1, and the IoU of matched labels per task and per class (only relevant for sidewalks and crosswalks). As in measuring time, we fit an LMM where normal and a GLMM where non-normal, of the same structure: F1/IoU ~ mode × expertise + order + set + (1 | participant). We correct per-class test p-values with Holm's sequential Bonferroni procedure [23], as above.

4.6.3 NASA Raw-TLX. We analyze the difference in NASA Raw-TLX scores between modes with Wilcoxon signed rank tests, with Holm's sequential Bonferroni procedure [23] applied to the individual scales.

4.6.4 Qualitative Analysis. For qualitative analysis of the participants' thinking-aloud and open-ended responses, we transcribed each user's session and analyzed them using a hybrid deductive-inductive coding approach [10]. We created an initial codebook based on the interview protocol and research questions, then two researchers coded three randomly selected interviews each. They then swapped and peer-checked each other's codes [43] and met to discuss disagreements and update the codebook accordingly. One researcher then coded all remaining interviews.

Table 4. Mean and standard deviation of labeling time in minutes and IoU per class, by mode. Participants were slower in AI Mode for every class, but IoU values were slightly higher in AI Mode across crosswalks and sidewalks. IoU is omitted for curb ramps as point labels have no area.

	Time per-class				IoU per class			
Class	Manual Mean	Manual SD	AI Mean	AI SD	Manual Mean	Manual SD	AI Mean	AI SD
Sidewalk	15.50	3.13	19.00	5.18	0.63	0.08	0.68	0.04
Crosswalk	6.07	1.86	8.73	3.15	0.77	0.05	0.78	0.03
Curb Ramps	5.20	1.29	7.90	2.82				
Total	26.50	5.08	35.60	9.46	0.68	0.06	0.71	0.03

4.7 Results

We find that, while occlusions and temporal differences between images remain challenges, multiple views enabled users to orient themselves and create more informed labels. Opinions on AI Mode were mixed, with participants highlighting the challenges and increased burden of correcting existing labels. AI Mode was significantly slower than Manual Mode, however, the labels were more accurate against our ground truth. We present these quantitative and qualitative findings together below. Mean session duration (including tasks, tutorials, and questionnaires) was 107.16 minutes (*SD*=21.33). Complete statistical test results are provided in the Appendix. Quotes are lightly edited for concision.

4.7.1 Occlusion and Inference Across Views. Participants encountered challenges both within individual images and across views; we identify three common strategies to manage those obstacles: (1) inferring using "common sense" or lived experience, (2) orienting and synthesizing across images using in-image landmarks, and (3) doing the same, but using the built-in UI elements and interactive features. However, the effectiveness of such strategies depended on the type of obstacle being faced.

The largest noted challenge was occlusions; foliage, cars, shadows, and building overhangs all hindered visibility: *"when you have tree canopy, you really need to assume where the sidewalk is"* (E1), *"I'm doing my best guess as to whether the sidewalk is under the building [overhang]"* (E7). Sidewalks were the most impacted (*e.g.*, *"sidewalks were the hardest because they're weirdly shaped and covered with trees,"* N3), compared to crosswalks (*"the crosswalks were easier because they're more in the open. There's no obstruction blocking them,"* N5), and curb ramps (*e.g.*, *"whenever I saw the yellow [ADA tactile bumps], all I had to do was place a dot,"* N6).

Users exercised three common strategies, often in combination, to overcome occlusion challenges. First, "common sense" and lived experience informed user inferences: *"I feel like it's here by the cars [and] pretty thin"* (N4), *"I'm using common sense and experience of walking on the street and assuming that the sidewalk wouldn't just stop"* (N7). Second, users used the two different views to label more confidently. For example, *"this looks much narrower in the [aerial] than in the street-view, so I'm gonna have to do a bit of projection of where the sidewalk [might be] under this building"* (E7), *"I'm going to use what I know from the street-view to inform where I'm adding things"* (E4). Third, the UI itself served a similar purpose, in helping users orient themselves: *"I can't see [the object] at all [in this view], so I'm just using the blue lines [to match it]"* (N3).

However, strategies that synthesized across images failed when there were *differences* in the images (due to capture time): *"[in this image] it was renovated, so it's really hard to judge whether this part has a curb ramp or not"* (E5), *"the crosswalk got moved between the street-view and aerial—I didn't notice with the crosswalk but I'm noticing it with the curb ramps"* (E7), *"where are these bushes? I don't see these at all"* (E3, when trying to orient themselves using the foliage).

4.7.2 AI Mode's Impact on Speed. Participants spent more time on average labeling in AI Mode (35.60 min, *SD*=9.46) than Manual Mode (26.50 min, *SD*=5.08). See Figure 7. The Shapiro-Wilk test confirmed normality of time residuals (W = .966, p = .398), justifying our use of an LMM. Based on an analysis of variance on our LMM, we find this difference was **statistically significant**: ($F(1, 12) = 21.99$, $p = .0005$, $\eta_p^2 = .65$). This was reflected in the users comments that inspecting and correcting proposed labels often took longer than creating labels from scratch. Inspecting each AI-proposed label took time: *"I tended to check every single label the AI made"* (E5), *"[I had to] look closely to make sure that it was [highlighting] the right thing"* (N6). Users also felt that the process of correction was slow: *"for crosswalks, it definitely slowed me down because that's a very easy rectangle. With AI, I'm trying to use a brush tool to link all these little rectangles together"* (N7), *"I had to use the brush to fill in all the little gaps, and have to change how big the brush and erase was. It took more effort"* (N5). This effect was felt across all three classes: sidewalks ($\chi^2(1, N = 32) = 5.81$, $p = .016$), crosswalks ($F(1, 12) = 21.26$, $p = .0005$, $\eta_p^2 = .64$), and curb ramps ($F(1, 12) = 28.98$, $p =< .001$, $\eta_p^2 = .71$) were all significantly slower in AI Mode (see Table 4). In spite of this, participants became faster as they labeled, with the second task (*mean/SD*=27.2/5.96 min) being completed faster than the first (*mean/SD*=34.9/9.61 min), statistically significantly so ($F(1, 12) = 12.47$, $p = .019$, $\eta_p^2 = .44$), with one participant commenting *"[Task 2] was easier than [Task 1] because I was just more used to the interface"* (E6). Task order was included in the LMM; the significant mode effect therefore reflects the difference *after* adjustment for this confound.

4.7.3 AI Mode's Impact on Accuracy. Annotations produced in AI Mode had higher IoUs (0.713, *SD*=0.031) than those from Manual Mode (0.676, *SD*=0.062). (IoU was calculated for sidewalk and crosswalk labels; point curb ramps were excluded from this measure.) See Figure 7. The Shapiro-Wilk test confirmed normality (W = .970, p = .480), and an analysis of variance on our LMM shows **statistical significance**: ($F(1, 12) = 7.98$, $p = .015$, $\eta_p^2 = .40$). This improvement is inspite of the fact that AI labels were an imperfect starting point (AI proposals alone scoring a IoU=.63, see Table 1). However, F1 did not differ significantly between modes ($F(1, 12) = 2.21$, $p = .16$, $\eta_p^2 = .16$).

Participants differed in their opinions on the accuracy of the AI labels. Some found it imprecise and needing correction: *"it felt like I was doing a lot more editing, whereas if I had just done it myself I'd do it right the first time"* (N7), *"this is the problem, all of these vertices"* (N1). Others saw the opposite: *"the AI making more polygons than necessary is a feature, not a bug, because it was precise. A human isn't going to expend the energy to manually do something that precise with a mouse"* (E3). This suggests that increased accuracy may come from AI labels being more precise *where they are correct*, intermixed with the extra energy to inspect existing labels: *"because they [weren't my labels], I wanted to double, triple check its work"* (N6). Per-class IoU difference was statistically significant for sidewalks ($F(1, 12) = 10.27$, $p = .015$, $\eta_p^2 = .46$), but not crosswalks ($F(1, 12) = 0.96$, $p = .348$, $\eta_p^2 = .07$), suggesting the difference applied to more arbitrarily shaped objects. Image set also significantly affected aggregate IoU ($F(1, 14) = 7.56$, $p = .017$, $\eta_p^2 = .39$) and F1 ($F(1, 12) = 6.05$, $p = .03$, $\eta_p^2 = .34$), indicating variation between the ease-of-labeling of Set A and Set B despite deliberate matching. Image set was included in the LMM, however; the significant mode effect therefore reflects the difference *after* adjustment for this confound.

4.7.4 Perceived Value of AI Assistance. Participants were evenly divided on AI Mode: eight preferred AI Mode for labeling, including three non-experts and five experts. Some preferences depended on proposal quality or the ability to toggle assistance (see Table 2). Interestingly, four of those who preferred AI Mode were faster in Manual Mode.

Participants evaluated AI Mode considering the perceived involved effort, time cost, value-gain, and overall enjoyment. For example, E8 said *"I prefer the AI Mode because it's much easier for me to complete the work, even though I realized that I spent much more time with the AI,"* and N2 mentioned *"I feel like it cut out half the work for me."* Others, however,

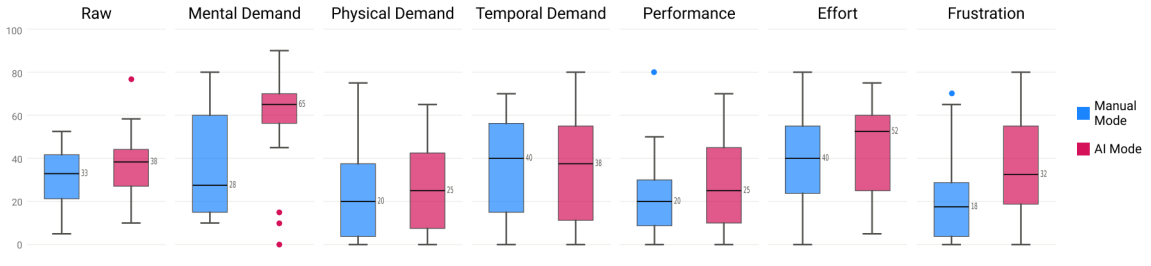


Fig. 8. Tukey box-and-whisker plots displaying NASA Raw-TLX scores by mode. Mental demand appears higher in AI Mode, but no scale was significantly different.

Table 5. We ask participants to complete the NASA Raw-TLX after completing both Manual and AI Mode tasks. We compare the scores using Wilcoxon signed-rank tests, with p-values for the individual scales (Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, Frustration) corrected with Holm's sequential Bonferroni procedure [23]. All results are non-significant, despite mental demand and frustration being higher in AI Mode.

Scale	Manual Median	Manual IQR	AI Median	AI IQR	Z	p
Mental Demand	27.50	45.00	65.00	13.75	-2.31	0.12
Physical Demand	25.31	33.75	26.56	35.00	-0.31	1.00
Temporal Demand	36.25	41.25	33.44	43.75	0.74	1.00
Performance	24.06	21.25	28.75	35.00	-0.76	1.00
Effort	38.13	31.25	45.00	35.00	-0.34	1.00
Frustration	17.50	25.00	32.50	36.25	-1.79	0.37
Raw	32.92	20.42	38.34	17.08	-1.37	0.18

found it frustrating: *"in AI Mode I just have to delete everything it puts in front of me,"* E2. Participants disagreed on whether having candidate labels at all was beneficial: *"it was helpful to see it pick up on something I wouldn't have even looked at"* (N6), *"[AI Mode] felt more leisurely, all I have to do is modify, not draw everything new"*, *"it's harder to analyze something you didn't create"* (E6). Several users couched their assessment of AI Mode based on the labeling expectations: *"if you just want something that works, either [AI or Manual Mode] works well. But for sophisticated stuff I would prefer manual"* (E8). Though, sometimes, the presence of AI *informed* the user what the expectation was: *"when doing it manually, I had a few moments where I didn't know if I should include [something], but then AI included it so I thought I should trust it and accept their suggestion"* (N5).

AI Mode was not significantly more strenuous than Manual Mode: neither overall workload nor individual dimensions of the NASA Raw-TLX scores (after Holm's sequential Bonferroni procedure [23]) were statistically significant (using Wilcoxon Signed-Rank tests), despite mental demand (Manual: *median*=27.5, *IQR*=45, AI Mode: *median*=65, *IQR*=13.75) appearing distinctly higher in AI Mode on the Tukey box-and-whisker plot (Figure 8). See Table 5 for Z-scores and p-values for each test.

As such, despite most users viewing the usefulness of AI Mode and the accuracy of AI labels, links positively (see Figure 9), users were still mixed on if they would use it to label "100 more image pairs," considering their own preferences for labeling and perceived cost *vs.* benefit.

4.7.5 Tool Interaction and Usability. Overall, the tool was well received (see Figure 9). The hotkeys, synchronous panning between street and aerial views, and the glowing link suggestions were particularly liked. However, some elements were found frustrating. One was the inability to pan while drawing a polygon. E7 was used to middle-mouse

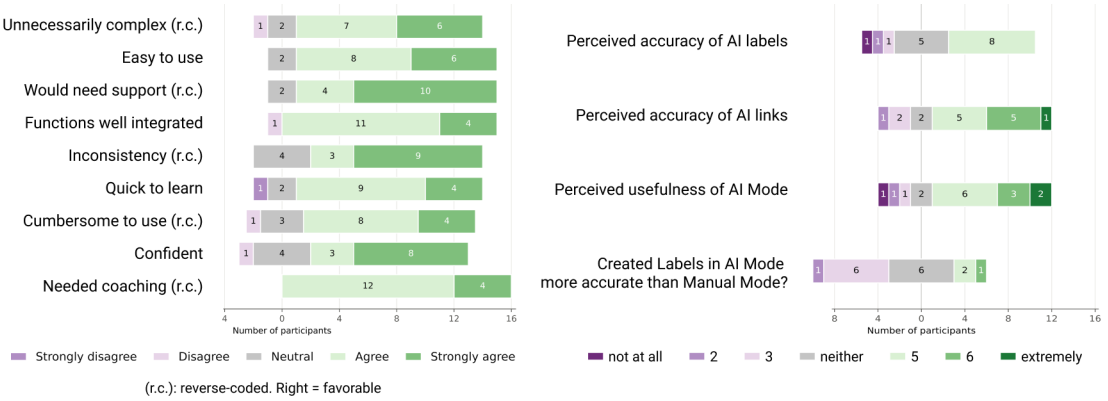


Fig. 9. Participants rated MultiMap's usability highly on the System Usability Scale (left). Likert ratings (7-point) of AI Mode (right) were more mixed, with participants rating their perceived accuracy of AI labels, links, and usefulness of AI Mode slightly above neutral. However, participant responses centered around neutral when asked if they believed their labels in AI Mode were more accurate than Manual Mode (when, in reality, labels created in AI Mode were more accurate).

panning from other software, which MultiMap avoids because it is inaccessible to trackpad users. The brush and erase tools were also frustrating, due to their imprecision relative to the polygon and box tools—the dominant tools used in AI Mode: *"with the brush you can't draw a straight line, so you need to freestyle"* (E8). Participants requested certain features. Several wished they *"could turn on the linking but turn off the label suggestions because I think the labels slowed me down"* (E6). Two participants wished they could pan beyond the 130° limit to match the aerial view's coverage, even though they were not required to label there: *"I didn't like being constrained in the imagery. I like being able to move around a little bit more"* (E2). Lastly, two participants suggested some form of feedback—either the system flagging classes the user had forgotten, or *"review and coaching [to make] one more confident in their performance"* (E1).

5 Discussion

We introduced and evaluated MultiMap, a multimodal streetscape annotation tool, to explore *cross-view* streetscape image annotation. In this section, we discuss key technical challenges and opportunities, design guidelines for future cross-view streetscape annotation tools and datasets, drawbacks of AI as a labeling guide, and the implications for future human-in-the-loop labeling systems.

5.1 Challenges and Opportunities for Cross-view Streetscape Annotation

A core goal of this research was to determine what challenges and opportunities are inherent in multimodal streetscape image annotation, and what considerations should be taken into account when both producing and consuming a cross-view streetscape image dataset. Results from our study indicate that there are certain unavoidable obstacles that can be ameliorated with multiple views, but not completely resolved. Occlusion is perhaps the biggest issue; common streetscape elements such as trees and shadows will make objects difficult or impossible to see. Transient objects, like cars, pedestrians, or micro-mobility devices also occlude lanes and sidewalks. Where *multimodality* can assist is in providing a different view to inform the overall scene. We found in our study that users often used what they could see in one image (*e.g.*, a sidewalk under an overhang) to inform their label in the other. Furthermore, we found that participants engaged in *landmarking* to orient themselves—trees, parking lanes, and crosswalks all helped users

determine the boundaries of objects. Future systems could take advantage of this sensemaking and provide more views of the same scene—different times (where there are fewer cars or shadows), seasons (so foliage is reduced), or a different camera angle could all inform and "fill in the gaps" of what is not visible in other images.

However, offering multiple views is not a panacea; it was a distraction for users when there were differences in the street structure in the images. Unfortunately, to avoid structural differences across multiple images would require the images to be taken at the same time—impractical when the largest sources of streetscape imagery are from third party platforms like open-source aerial imagery [24] and Google Street View [6]. This can be combated by using datasets that were produced around similar times, but permanent or transient changes in the streetscape are not completely avoidable. This is also in tension with the above proposal (to show images at different times and seasons).

As such, streetscape dataset producers and consumers must consider (1) to what degree the images and features of interest will be occluded, (2) how vulnerable the data sources are to temporal differences causing structural inconsistency, and (3) what level of annotator inference (*i.e.*, inferring where an object should be even though it is occluded) is tolerable. Given that occlusion and temporal differences are unavoidable, future work should explore to what degree computer vision approaches are robust to temporal and structural differences in multimodal imagery.

5.2 Benefits and Drawbacks of AI-Assisted Labeling

Prior to our study, we hypothesized that AI-assisted labeling would be faster and more accurate, echoing past work [2, 3, 13, 36, 38]. For MultiMap, however, accuracy improved, but labeling was slower, and users were divided on the AI's perceived utility. These findings suggest that AI assistance introduces a tradeoff: improvements in label quality must be weighed against the time and effort required to obtain them, and that these benefits may not always be clear to the user.

Our results suggest that the accuracy gain may not be independent of the increased time cost. Labels were more accurate, but participants also spent longer on them, with some calling this phenomenon out directly: *"AI Mode forced me to think a little bit more critically. It might encourage you to be a little bit more careful and go a little slower"* (E6). As such, we cannot distinguish whether the increased accuracy arising from AI Mode was due to the AI's proposed labels, the induced, more deliberate human labeling, or both—but, regardless, with AI assistance, labels *were* more accurate.

However, participants did not consistently perceive the measured accuracy improvement. When asked how much better they felt their labels were in AI Mode on a scale of one to seven, the median response was 4 (see Figure 9). This meant users were not clearly aware of any benefit of using AI assistance, only perceiving the time cost: *"it would be quicker to do it myself"* (N5). These responses suggest that, for relatively simple operations, reviewing and correcting an AI proposal can feel more burdensome than creating labels manually. In contrast, participants generally found AI *linking* useful, suggesting that users felt that linking as a task was one where the AI saved them more effort than it cost, perhaps due to the increased cognitive load of matching between image views.

These observations suggest that an AI-propose, human-correct system should be designed around the demands of the labeling tasks and the users' and dataset creators' priorities. Future systems should evaluate accuracy, completion time, and AI correction effort to determine whether the assistance actually offers a worthwhile tradeoff over manual labeling. Additionally, if the AI assistance's benefits (in MultiMap's case, increased accuracy) are not made clear to the user, they may deem the AI not worth the effort and choose to go without it, sacrificing any gains the AI assistance provides. Systems should make otherwise difficult-to-observe quality gains visible *alongside* the time and correction effort to achieve them, allowing both users and dataset creators to decide when AI assistance is worthwhile.

5.3 Implications for Design

Drawing on our study findings, we synthesize five key recommendations for the design of multimodal annotation tools, with application beyond streetscape imagery.

- **Annotation Expectations** Participants indicated their labeling behavior would depend on *expectations*, namely *precision* (*e.g.*, how tightly the annotation must hug the object) and *definition* (*e.g.*, what constitutes a curb ramp). Not only do labeling expectations influence dataset creation, but also what downstream consumers should expect from the dataset. Dataset producers should be clear about *what* is labeled and in *what* way.
- **Synchronicity** Participants appreciated and utilized MultiMap's interaction synchronicity (*e.g.*, panning and zooming). Future multimodal annotation systems should embed as much synchronicity as possible to reduce mental burden for the user, even if the algorithm is coarsely implemented (*e.g.*, our automated linking algorithm).
- **Feedback** Labeling systems should offer a way for the user to understand the quality of their performance, and if they are "on track," in the form of coaching or real-time corrective guides [38]. Conversely, users should be able to *provide* feedback to indicate when the two images' structure is different due to temporal differences.
- **Personalization** Users of MultiMap requested personalization, including toggling individual AI Mode features, or changing the toolbar hotkeys. Labeling systems should allow customization and personalization to give users the agency of determining (and only activating) features they find useful.

5.4 Limitations

Our work has seven primary limitations. First, we performed our study with only three different infrastructure types, in only one U.S. city. Future work should expand beyond a U.S. focus and our three classes. Second, to avoid fatigue, each participant labeled only 20 image pairs, so our conclusions may not extend to longer, more dedicated labeling sessions. Future work should explore these interactions in a larger deployment study. Third, we only provided users rough guidelines for labeling, to explore how each participant would make their own judgments (emulating an environment with little researcher supervision). As such, accuracy error could potentially come from mismatched labeling expectations. Future work should explore whether AI assistance improves accuracy given strict labeling criteria. Fourth, the leaderboard in MultiMap implemented to encourage participants to continue the study may have incentivized participants to label faster, potentially interfering with our analysis. Fifth, we do not perform a comprehensive analysis of where Nano Banana 2's cleanup and detection succeeded and failed, and in what ways. Sixth, our results show that AI-assisted labels were more accurate—however, this difference is small (0.71 *vs.* 0.68 IoU). This difference may not be meaningful depending on context. Seventh, MultiMap only operates on two views: one street level and one aerial. Future work should explore more and different camera angles, including non-panoramic images, drone footage, and greater variation in time/season.

6 Conclusion

MultiMap marks the first study exploring the potential benefits, drawbacks, and challenges of multimodal streetscape annotation—a rich avenue for future, more powerful computer vision approaches that can leverage multiple perspectives. We contribute: **formative findings** on the considerations necessary for multimodal streetscape annotation, **design guidelines** for future systems in this space, **a novel MLLM-powered pipeline** to detect streetscape infrastructure in street-view and aerial imagery, and **formative findings** on AI-assisted labeling systems. Finally, we will release MultiMap's code for open science upon acceptance.

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A Appendix

A.1 Prompts

A.1.1 Cleaning Aerial Images. *"This is an aerial image of a street. Remove any trees, shadows, and cars occluding the road and sidewalks, but do not create objects that don't already visibly exist."*

A.1.2 Cleaning Street-level Images. *"This is an image of a street. Remove all cars and pedestrians from the image, but do not create objects that don't already visibly exist."*

A.1.3 Increasing Paint and Road Marking Visibility. *"Make road markings more clear in this image. Do NOT alter existing geometry or the marking types, only fill in lanes or lines that are messy or occluded."*

A.1.4 Detecting Objects. *"Generate a semantic segmentation visualization image, using this color mapping: {"sidewalk": "green", "road": "red", "crosswalks": "blue", "buildings": purple}. Crosswalks are the entire area of the crosswalk, not just the stripes."*

A.1.5 Detecting Curb Ramps. *"Draw a circular marker in pink on top of each curb ramp in this image. Each individual curb ramp should get its own marker."*

A.2 Full Test Results

Table 6. Shapiro-Wilk tests of normality for time, IoU, and F1-Score, aggregated per-class and in Total.

DV	W	p	normal?
Total Time	0.97	0.40	yes
Sidewalk Time	0.90	0.01	no
Crosswalk Time	0.98	0.84	yes
Curb Ramp Time	0.99	0.94	yes
IoU (pooled)	0.97	0.48	yes
Sidewalk IoU	0.97	0.58	yes
Crosswalk IoU	0.96727	0.4277	yes
F1 (pooled)	0.96275	0.326	yes
Sidewalk F1	0.95332	0.1789	yes
Crosswalk F1	0.97014	0.5033	yes
Curb ramp F1	0.9626	0.3231	yes

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Table 7. Full statistical test results for Mode (Manual *vs.* AI), task order, image set, and Expertise on time and accuracy.

DV	manual, mean (SD)	ai, mean (SD)	test	p	holm_p	significance	η_p^2
Mode							
Total task time	26.53 (5.08)	35.59 (9.46)	F(1, 12.00) = 21.99	<.001		***	0.65
Sidewalk time	15.54 (3.13)	19.00 (5.18)	χ^2(1, N=32) = 5.81	0.02	0.02	*	
Crosswalk time	6.07 (1.87)	8.73 (3.15)	F(1, 12.00) = 21.26	<.001	0.001	***	0.64
Curb-ramp time	5.20 (1.29)	7.90 (2.82)	F(1, 12.00) = 28.98	<.001	<.001	***	0.71
IoU (pooled)	0.676 (0.062)	0.713 (0.031)	F(1, 12.00) = 7.98	0.02		*	0.40
Sidewalk IoU	0.629 (0.076)	0.679 (0.035)	F(1, 12.00) = 10.27	0.01	0.02	*	0.46
Crosswalk IoU	0.765 (0.054)	0.778 (0.034)	F(1, 12.00) = 0.96	0.35	0.35		0.07
F1 (pooled)	0.936 (0.019)	0.944 (0.013)	F(1, 12.00) = 2.21	0.16			0.16
Sidewalk F1	0.963 (0.031)	0.978 (0.021)	F(1, 12.00) = 2.29	0.16	0.47		0.08
Crosswalk F1	0.969 (0.029)	0.965 (0.032)	F(1, 12.00) = 0.33	0.58	1.00		0.03
Curb-ramp F1	0.879 (0.053)	0.885 (0.048)	F(1, 12.00) = 0.29	0.60	1.00		0.02
Order							
Total task time	34.89 (9.61)	27.23 (5.96)	F(1, 12.00) = 16.66	0.002		**	0.58
Sidewalk time	19.06 (4.57)	15.48 (3.91)	χ^2(1, N=32) = 6.41	0.01	0.02	*	
Crosswalk time	8.41 (3.45)	6.39 (1.77)	F(1, 12.00) = 9.51	0.01	0.02	*	0.44
Curb-ramp time	7.45 (2.95)	5.64 (1.74)	F(1, 12.00) = 12.47	0.004	0.01	*	0.51
IoU (pooled)	0.700 (0.048)	0.690 (0.057)	F(1, 12.00) = 0.60	0.45			0.05
Sidewalk IoU	0.663 (0.063)	0.645 (0.065)	F(1, 12.00) = 1.32	0.27	0.55		0.10
Crosswalk IoU	0.769 (0.038)	0.774 (0.052)	F(1, 12.00) = 0.15	0.71	0.71		0.01
F1 (pooled)	0.938 (0.018)	0.942 (0.016)	F(1, 12.00) = 0.42	0.53			0.03
Sidewalk F1	0.974 (0.023)	0.966 (0.031)	F(1, 12.00) = 0.71	0.42	0.72		0.03
Crosswalk F1	0.964 (0.032)	0.970 (0.029)	F(1, 12.00) = 0.91	0.36	0.72		0.07
Curb-ramp F1	0.874 (0.047)	0.891 (0.054)	F(1, 12.00) = 2.28	0.16	0.47		0.16
Set							
Total task time	30.50 (9.16)	31.62 (8.64)	F(1, 12.00) = 0.50	0.50			0.04
Sidewalk time	16.80 (3.84)	17.74 (5.27)	χ^2(1, N=32) = 0.17	0.68	1.00		
Crosswalk time	7.60 (3.18)	7.20 (2.64)	F(1, 12.00) = 0.43	0.52	1.00		0.03
Curb-ramp time	6.40 (2.96)	6.70 (2.16)	F(1, 12.00) = 1.28	0.28	0.84		0.10
IoU (pooled)	0.712 (0.043)	0.677 (0.055)	F(1, 12.00) = 7.56	0.02		*	0.39
Sidewalk IoU	0.676 (0.047)	0.632 (0.071)	F(1, 12.00) = 8.20	0.01	0.03	*	0.41
Crosswalk IoU	0.780 (0.049)	0.763 (0.040)	F(1, 12.00) = 1.43	0.26	0.26		0.11
F1 (pooled)	0.947 (0.015)	0.934 (0.017)	F(1, 12.00) = 6.05	0.03		*	0.34
Sidewalk F1	0.973 (0.020)	0.968 (0.034)	F(1, 12.00) = 0.28	0.61	0.61		0.01
Crosswalk F1	0.953 (0.027)	0.980 (0.027)	F(1, 12.00) = 17.20	0.001	0.003	**	0.59
Curb-ramp F1	0.907 (0.048)	0.857 (0.040)	F(1, 12.00) = 20.96	<.001	0.002	**	0.64
Expertise							
Total task time	29.90 (6.22)	32.22 (4.87)	F(1, 14.00) = 1.12	0.31			0.07
Sidewalk time	16.39 (3.78)	18.15 (2.35)	χ^2(1, N=32) = 1.56	0.21	0.64		N/A
Crosswalk time	7.50 (2.15)	7.30 (2.03)	F(1, 14.00) = 0.04	0.85	1.00		0.003
Curb-ramp time	6.31 (1.82)	6.79 (1.77)	F(1, 14.00) = 0.34	0.57	1.00		0.02
IoU (pooled)	0.686 (0.036)	0.703 (0.042)	F(1, 14.00) = 0.69	0.42			0.05
Sidewalk IoU	0.647 (0.041)	0.661 (0.052)	F(1, 14.00) = 0.33	0.58	0.58		0.02
Crosswalk IoU	0.761 (0.034)	0.782 (0.039)	F(1, 14.00) = 1.34	0.27	0.53		0.09
F1 (pooled)	0.939 (0.015)	0.941 (0.009)	F(1, 14.00) = 0.18	0.68			0.01
Sidewalk F1	0.973 (0.013)	0.967 (0.022)	F(1, 14.00) = 0.37	0.55	1.00		0.01
Crosswalk F1	0.965 (0.025)	0.968 (0.020)	F(1, 14.00) = 0.08	0.78	1.00		0.01
Curb-ramp F1	0.876 (0.042)	0.888 (0.031)	F(1, 14.00) = 0.40	0.54	1.00		0.03

Table 8. Full statistical test results for the Mode:Expertise interaction.

DV	Non-expert manual, mean (SD), n=8	Non-expert AI, mean (SD), n=8	Expert manual, mean (SD), n=8	Expert AI, mean (SD), n=8	test	p	holm_p	sig	η_p^2
mode:expertise									
Total task time	24.64 (3.50)	35.16 (11.27)	28.42 (5.91)	36.01 (8.03)	F(1, 12.00) = 0.47	0.51			0.04
Sidewalk time	14.13 (2.42)	18.66 (6.42)	16.96 (3.26)	19.33 (4.00)	χ^2(1, N=32) = 0.68	0.41	1.00		N/A
Crosswalk time	5.98 (1.37)	9.02 (3.68)	6.17 (2.37)	8.43 (2.74)	F(1, 12.00) = 0.12	0.74	1.00		0.01
Curb-ramp time	5.09 (0.76)	7.53 (3.29)	5.31 (1.72)	8.27 (2.43)	F(1, 12.00) = 0.96	0.35	1.00		0.07
IoU (pooled)	0.664 (0.064)	0.709 (0.024)	0.689 (0.061)	0.717 (0.038)	F(1, 12.00) = 0.47	0.51			0.04
Sidewalk IoU	0.617 (0.071)	0.677 (0.025)	0.641 (0.084)	0.681 (0.045)	F(1, 12.00) = 0.39	0.54	1.00		0.03
Crosswalk IoU	0.751 (0.063)	0.771 (0.026)	0.778 (0.044)	0.786 (0.041)	F(1, 12.00) = 0.21	0.65	1.00		0.02
F1 (pooled)	0.934 (0.021)	0.943 (0.017)	0.938 (0.019)	0.945 (0.009)	F(1, 12.00) = 0.04	0.84			0.00
Sidewalk F1	0.969 (0.027)	0.977 (0.022)	0.956 (0.036)	0.978 (0.021)	F(1, 12.00) = 0.57	0.47	0.47		0.02
Crosswalk F1	0.976 (0.031)	0.954 (0.030)	0.961 (0.027)	0.975 (0.032)	F(1, 12.00) = 7.35	0.02	0.06		0.38
Curb-ramp F1	0.861 (0.055)	0.891 (0.062)	0.897 (0.049)	0.879 (0.034)	F(1, 12.00) = 4.77	0.05	0.10		0.28